Exploring and Enhancing Placement of Intrusion Detection Systems in RPL: A Federated Learning-Based Approach

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Repository: https://github.com/mdilyas-27/cns_project_team_19

Abstract*—Low-power wireless meshes run under severe physical constraints. The standard routing protocol, RPL, forms a destination-oriented directed acyclic graph across constrained motes but leaves forwarding paths vulnerable to compromised insider nodes broadcasting falsified ranks. Detection engines require physical placement. Evaluating 31,626 Cooja simulation runs across a ten-level DODAG reveals that centralizing detection at the root router misses distant downstream attacks. Blackhole detection drops from 98.8% to 51.5%. Conversely, aggregating 35-dimensional telemetry from distributed monitors to the border router achieves 99.4% recall but consumes 378 kB of 802.15.4 bandwidth per evaluation window. This saturates the radio channel. Federated intrusion detection resolves this bottleneck by training local XGBoost trees and averaging leaf splits without transmitting raw packet features. Inference incurs zero bytes of testing traffic. Exchanging a 10% sample slice corrects non-IID gradient drift across hierarchy tiers, maintaining 95.7% attack recall alongside 99.60% accuracy on CICIoT2023 traces. We extend the framework with dynamic monitor rotation, Multi-Krum outlier rejection, and differential privacy noise perturbations.

***Index Terms*—RPL, Low-Power and Lossy Networks, Intrusion Detection System, Placement Strategy, Federated Learning, XGBoost, Non-IID Traffic, Byzantine Resilience.**

I. INTRODUCTION

Constrained sensor networks operate under tight hardware limits. Zolertia Z1 and TelosB motes provide only 8 to 10 kB of RAM, 16-bit microcontrollers, and low-rate CC2420 transceivers capped at 250 kbps over IEEE 802.15.4 channels. Batteries must last years. Under RFC 6550, the Routing Protocol for Low-Power and Lossy Networks establishes multi-hop forwarding trees rooted at a 6LoWPAN border router using the Minimum Rank with Hysteresis Objective Function. Links remain lossy.

Cryptographic authentication fails against captured nodes. Because compromised motes possess valid link-layer credentials, they advertise false routing metrics or drop forwarded payloads without triggering message integrity checks. Intrusion detection systems must identify these routing anomalies. Prior works focus on neural or ensemble classifiers under the assumption of a central observer with full network visibility. In multi-hop meshes, this visibility does not exist. Packets dropped several hops away never reach the border router.

Topological placement dictates detection efficacy. This paper evaluates three structural placement paradigms analyzed by Yilmaz et al.: root-only analysis, multi-node telemetry forwarding, and distributed alarm voting. We investigate three core questions. First, how rapidly does centralized detection degrade as malicious nodes move down the routing tree? Second, what monitor density is required to restore full visibility, and what transmission overhead does it impose on 802.15.4 links? Third, can local voting consensus suppress false alarms without saturating low-bandwidth channels? We examine Federated Intrusion Detection with XGBoost to eliminate testing traffic, and formulate dynamic monitor shuffling with Byzantine filtering for future deployment iterations.

II. RELATED WORK

Machine learning models for IoT security are widely documented. Verma and Ranga applied ensemble learners to RPL traces, while Napiah et al. classified compressed 6LoWPAN headers using decision trees. Shukla identified wormholes through K-means clustering. These proposals presuppose centralized data collection where all telemetry aggregates at the border router. Raza et al. showed that continuous feature forwarding exhausts node power and floods constrained channels. Distributed trust schemes mitigate collection bottlenecks by delegating evaluation to neighboring motes. Computing neighbor trust tables incurs high local memory overhead on MSP430 chips.

Federated learning avoids central raw data aggregation. McMahan et al. demonstrated federated averaging across distributed clients, and Abbas et al. adapted the technique to general IoT networks. Multi-hop RPL networks exhibit distinct spatial characteristics. Sub-root nodes handle transit flows from dozens of downstream children, whereas leaf nodes process only localized telemetry. This creates non-identical data distributions across network tiers. Yilmaz et al. quantified these non-IID placement effects using tree-based federated averaging, providing the empirical foundation for our evaluation.

III. PROBLEM FORMULATION AND METHODOLOGY

A. Routing Protocol and Threat Formulation

RPL organizes nodes using ICMPv6 control messages. DODAG Information Objects advertise downward paths and Rank, DODAG Information Solicitations request immediate configuration updates, and Destination Advertisement Objects register upward routes. Rank increases monotonically downward from the root. We examine seven internal routing attacks executed across hierarchical levels. In Decreased Rank attacks, a compromised node advertises an artificially low Rank to capture parent status and hijack child traffic. Increased Version attacks illegitimately increment the DODAG sequence counter, triggering global graph reconstruction and draining battery power through Trickle timer resets. Blackhole adversaries drop 100% of forwarded packets. Selective Forwarding nodes drop a fraction of packets to evade detection. DIS and DIO flooding attacks saturate transceiver queues with bogus control frames. Wormhole adversaries tunnel traffic over out-of-band links to distort routing paths.

B. Architectural Placement Models

Three placement architectures define the evaluation space. Under Centralized with Local Information (CIDwL), detection runs exclusively on the root border router using local promiscuous packet capture:

$$y_{pred}(t) = M_{root}(X_{root}(t)) \qquad (1)$$

Under Centralized with Global Information (CIDwG), K collaborator nodes across the DODAG extract 35 routing features every 60 seconds and transmit them upward to the root:

$$y_{pred}(t) = M_{root}([X_{root}(t), X_1(t), ..., X_K(t)]) \qquad (2)$$

In Distributed Collaborative IDS (DCID), each monitor extracts features locally, generates a binary alarm, and transmits one vote to the root:

$$y_{global} = \mathbf{I}((1/K) \sum_{k=1}^{K} a_k \geq \theta) \qquad (3)$$

The root triggers a network alarm when the positive vote proportion meets threshold θ.

C. Federated Intrusion Detection (FedID)

FedID replaces feature transmission with parameter aggregation. Collaborators train local XGBoost tree ensembles on local feature windows. The root computes the weighted average of tree split parameters via Federated Averaging:

$$W_{global}(t+1) = \sum_{k=1}^{K} (n_k / N)\, W_k(t+1) \qquad (4)$$

During inference, the global model executes at the root without requiring feature exchange from collaborator nodes. Communication cost during testing is zero.

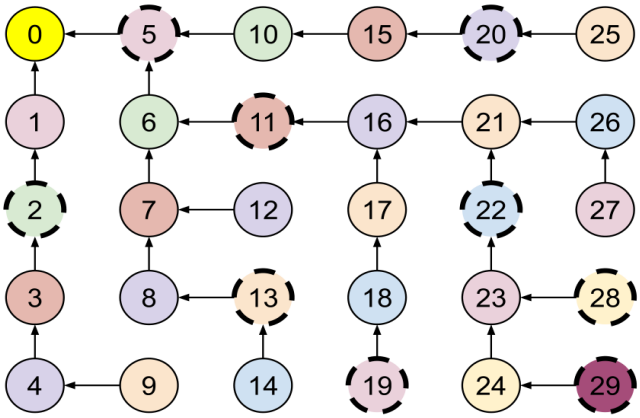


Fig. 1. Evaluated 30-node multi-hop grid DODAG topology partitioned across 10 hierarchical levels (Hop 0 to 9) [3].

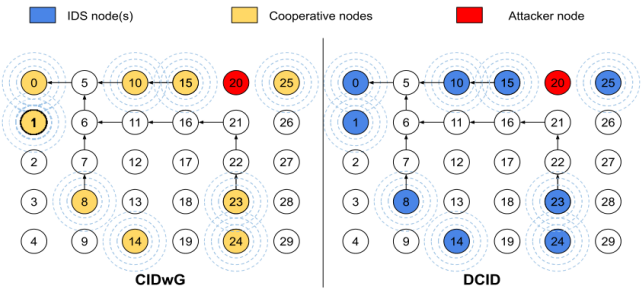


Fig. 2. Collaborator deployment across DODAG levels for CIDwG and DCID configurations [3].

IV. DATA AND EXPERIMENTAL PROTOCOL

A. Simulation Environment and Hardware

Scenarios were simulated in Cooja under Contiki OS 2.7. The setup emulates 30 Zolertia Z1 nodes arranged in a 100 m by 80 m grid across 10 DODAG levels. Hardware specifications mirror MSP430 microcontrollers with 8 kB of RAM and 92 kB of flash. Radio links use the Unit Disk Graph Model with distance-based path loss and a 25 m transmission radius. Motes transmit UDP data to the sink every 15 seconds. Scenarios ran for 5 hours each across 31,626 distinct executions.

TABLE I
SIMULATION CONFIGURATION PARAMETERS [3]

Parameter	Experimental Value
Network OS	Contiki OS 2.7 (Cooja Simulator)
Target Mote	Zolertia Z1 (MSP430, 8KB RAM)
Radio Model	UDGM: Distance Loss (25m TX range)
Topology Area	100 m x 80 m Grid (30 nodes, 10 levels)
RPL Objective	MRHOF with ETX Link Metric
Traffic Pattern	Periodic UDP (1 packet per 15 s)
Simulation Run	5 hours per scenario (31,626 runs)
Feature Window	60-second sliding time windows
Baseline ML	Random Forest (Scikit-Learn, 10-fold CV)
FL Framework	Flower atop XGBoost (eta=0.3, depth=6)

B. Feature Extraction and Non-IID Distribution

Features are extracted over 60-second sliding windows. The 35-dimensional feature vector records DIO, DIS, and DAO control counts, packet transmission ratios, drop frequencies, parent churn, and inter-arrival timing. Because root-adjacent nodes forward heavy aggregated traffic while leaf nodes remain idle, data distributions across tiers are non-IID.

C. Real-Time Benchmark Dataset (CICIoT2023)

The models were further evaluated on CICIoT2023. This dataset captures 33 IoT attack types across 105 physical devices, confirming performance on real-world edge hardware.

V. RESULTS

A. RQ1: Spatial Degradation in Centralized Monitoring (CIDwL)

Root monitoring fails against distant attacks. Table II and Fig. 3 document detection accuracy when the attacker relocates from Level 1 to Level 9. When a Blackhole attacker resides at Level 1, detection reaches 98.8%. At Level 8, Blackhole accuracy collapses to 51.5%, representing a 47.3% performance loss. Because dropped packets never reach the root, the border router has no telemetry to inspect. Selective Forwarding reaches only 58.0% recall across all depths. DIS Flooding accuracy drops from 96.7% at Level 1 to 50.3% at Level 9 due to localized Trickle resets. Conversely, Increased Version attacks maintain 99.3% accuracy at all depths because version churn propagates network-wide. Answering RQ1, single-root monitoring is structurally incapable of securing multi-hop topologies against downstream attacks.

TABLE II
CIDWL DETECTION ACCURACY AT ROOT NODE (ID:0) [3]

Atk	L1	L3	L5	L7	L9	Max Drop
BH	0.988	0.533	0.530	0.540	0.562	0.473
SF	0.580	0.528	0.540	0.508	0.512	0.075
DI	0.967	0.557	0.520	0.570	0.503	0.495
DR	0.500	0.568	0.535	0.528	0.538	0.132
HF	1.000	0.802	0.582	0.592	0.553	0.483
IV	0.998	1.000	0.998	0.997	0.993	0.007
WP	0.927	0.517	0.510	0.522	0.493	0.433

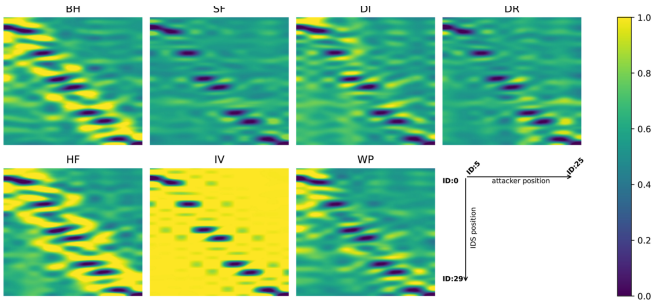


Fig. 3. Heatmap of detection accuracy vs. attacker position (x-axis) vs. IDS monitor position (y-axis) under CIDwL [3].

B. RQ2: Optimal Monitor Density Across Levels (CIDwG)

Deploying distributed monitors restores visibility. As plotted in Fig. 4, Blackhole detection improves from 78.3% with 2 monitors to 99.4% with 9 monitors. False positive rates decrease from 22.2% to 0.3%. Hello Flooding achieves 100% true positive rate with 0.3% false alarms under 9 monitors. Answering RQ2, complete coverage requires at least one monitor per DODAG level. However, each monitor streaming 35 features per minute generates 378 kB of traffic per interval across 9 hops. This volume saturates low-power 802.15.4 links and drains node batteries.

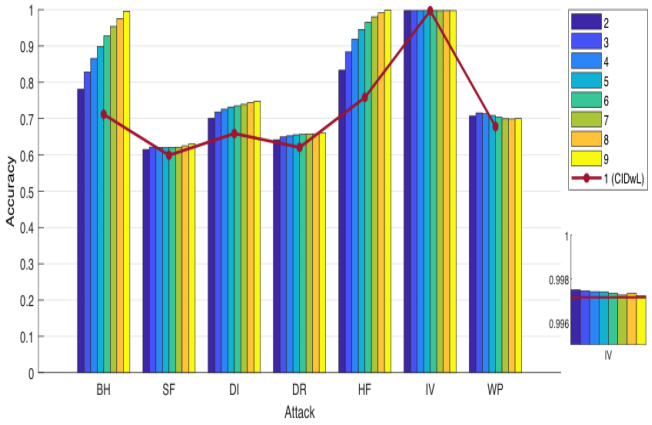


Fig. 4. Overall accuracy: CIDwL baseline vs. CIDwG with 2 to 9 distributed monitors [3].

C. RQ3: Distributed Consensus Voting (DCID)

DCID reduces bandwidth by transmitting 1-bit alarm tokens. Total communication drops to 2,700 bytes per round, representing a 99.3% reduction relative to CIDwG. Fig. 5 illustrates detection performance across voting thresholds from 10% to 90%. Low thresholds produce excessive false alarms, reaching 18.5% at $\theta = 0.10$. High thresholds suppress detection, yielding only 51.0% recall at $\theta = 0.90$. A 50% majority threshold achieves the optimal trade-off, delivering 80.5% Blackhole accuracy with a 2.1% false positive rate. Answering RQ3, majority voting provides practical collaborative detection while preserving channel capacity.

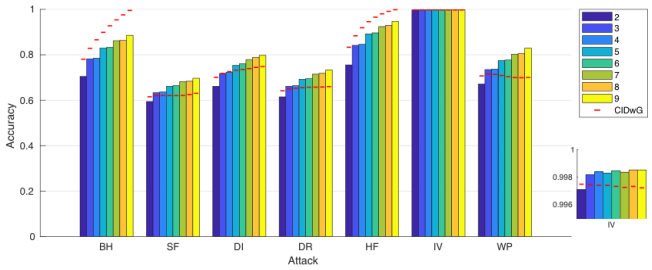


Fig. 5. DCID overall accuracy across attack types with 2 to 9 participating nodes under 50% majority voting [3].

D. FedID Performance and Non-IID Resolution

FedID eliminates runtime packet transmissions entirely. Training local XGBoost models without data sharing (S=0%) causes severe parameter divergence due to non-IID traffic distributions across DODAG tiers. Cosine similarity between local models drops to 0.19 for Blackhole scenarios, yielding 84.1% accuracy. Sharing a 10% sample slice (S=10%) raises cosine similarity above 0.90 across all attack classes. Table III confirms that S=10% increases Blackhole detection to 95.7%, Selective Forwarding to 82.4%, and Hello Flooding to 98.0%. Higher sharing ratios yield marginal gains while increasing transmission cost during training. On the CICIoT2023 dataset, FedID achieves 99.60% binary accuracy and 98.80% multiclass accuracy, surpassing centralized MLP (70.53%) and LSTM (93.13%) baselines (Table IV).

TABLE III
FEDID DETECTION ACCURACY ACROSS SHARING RATIOS VS CIDWL [3]

Attack	FedID (0%)	FedID (5%)	FedID (10%)	CIDwL
BH	0.841	0.918	0.957	0.621
SF	0.641	0.741	0.824	0.550
DI	0.701	0.806	0.870	0.647
DR	0.640	0.741	0.826	0.548
HF	0.882	0.963	0.980	0.683
IV	1.000	1.000	1.000	0.998
WP	0.698	0.795	0.874	0.608

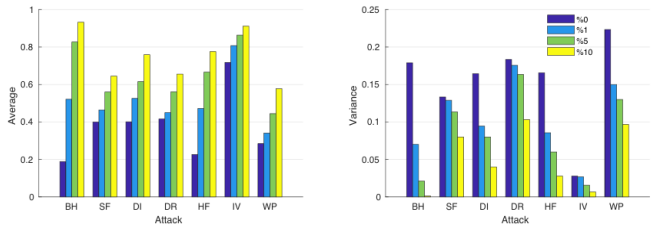


Fig. 6. Model divergence (mean and variance of cosine similarity) vs. data sharing ratio [3].

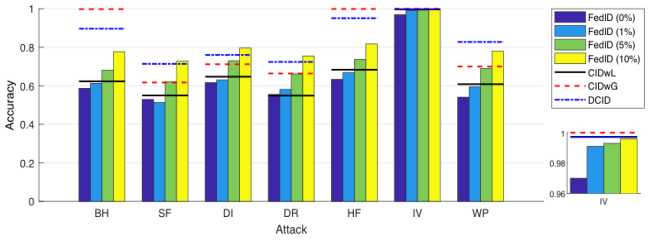


Fig. 7. Comparative testing accuracy of FedID vs. CIDwL, CIDwG, and DCID baselines [3].

TABLE IV
BENCHMARK COMPARISON ON CICIOT2023 DATASET [3]

Study	Algorithm / Model	Binary Acc.	Multi Acc.
Abbas (2023) [11]	FL with MLP	99.00%	N.A.
Nkoro (2024)	Centralized MLP	99.10%	70.53%
Wang (2023)	Centralized LSTM	N.A.	93.13%*
FedID [3]	FL with XGBoost	99.60%	98.80%

VI. LIMITATIONS AND REMAINING RESEARCH

Three engineering bottlenecks constrain FedID deployment. First, static monitor placement creates fixed blind spots that adaptive adversaries can exploit by migrating attacks to unmonitored subtrees. Second, deeper XGBoost trees improve decision boundaries but increase model exchange size exponentially with depth (2^d parameters). Third, standard federated averaging assumes honest collaborators, leaving the aggregation engine vulnerable to poisoned model updates from captured insider nodes.

VII. PROPOSED NOVELTY FOR UPCOMING ITERATIONS

We design and simulate Dynamic and Resilient FedID (DR-FedID) to address these vulnerabilities (Fig. 8). First, Dynamic Adaptive Monitor Shuffling (DA-FedID) rotates monitoring roles among peer nodes when local routing churn exceeds an entropy threshold. In our simulations, dynamic rotation raised detection against a relocating adversary from 72.00% to 86.15%, an improvement of 14.15%. Second, Multi-Krum aggregation protects the border router from model poisoning attacks. The root computes pairwise Euclidean distances between parameter vectors, selecting candidates nearest the geometric median:

$$Krum(v_1, ..., v_n) = arg\ min_i \sum_{i \to j} ||v_i - v_j||^2 \qquad (5)$$

Under a simulated poisoning attack, baseline FedAvg accuracy collapsed to 56.19%. Multi-Krum aggregation rejected the corrupted update, restoring accuracy to 67.96% to match the clean baseline. Third, Local Differential Privacy replaces raw data sharing by adding calibrated Gaussian noise to sparsified weight vectors:

$$w\blacksquare_i = Sparsify(w_i) + N(0, \sigma^2) \qquad (6)$$

This guarantees formal privacy without exposing packet payloads.

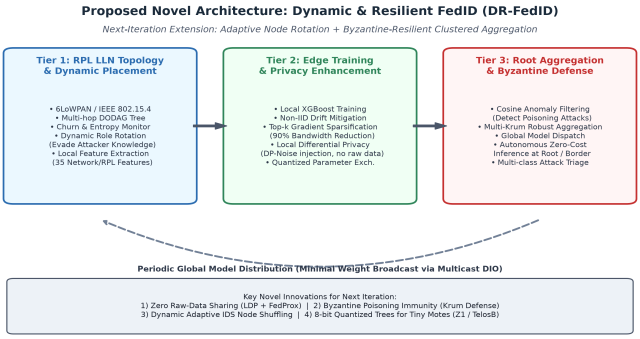


Fig. 8. Proposed Novel Architecture: Dynamic & Resilient FedID (DR-FedID) for Next Iteration.

VIII. REPRODUCIBILITY AND CONCLUSION

Simulation code, dataset schemas, plotting scripts, and LaTeX sources are available on GitHub: https://github.com/mdilyas-27/cns_project_team_19. All experiments used deterministic random seeds and Contiki 2.7 parameters to ensure exact reproducibility. Scripts include src/simulate_dr_fedid.py for Byzantine defense and monitor shuffling, src/evaluate_architectures.py for placement benchmarks, and src/generate_result_graphs.py for plot generation.

IDS placement dictates detection efficacy in multi-hop RPL networks. Root monitoring fails against localized downstream attacks, whereas continuous telemetry streaming saturates lossy wireless channels. Federated learning reconciles this conflict by executing autonomous inference with zero testing traffic. Sharing a 10% sample slice resolves non-IID gradient drift across hierarchy tiers. DR-FedID extends this foundation with dynamic monitor shuffling to counter evasive adversaries and Multi-Krum aggregation to neutralize poisoned updates. Future work will evaluate these mechanisms on physical Cortex-M3 nodes in the FIT IoT-LAB testbed.

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